

General-purpose crawler unit



CuGoMEGA M2

Base kit with options

Assembly manual



Cubi
CRAWL YRo Solo RFID

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1 Safety Precautions

	Warning This section indicates items that could result in death or injury to users.
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	Warning
When performing assembly or similar tasks, disconnect the power. Doing it with the power on may cause accidental operation and could lead to an accident. Do not touch moving parts such as tracks and chains during power-on or operation. Hands or fingers may get caught and suffer serious injuries. The machine has sharp edges on aluminum frame corners and other areas. Exercise caution during assembly and transport.	

2 Product Configuration

2.1 CuGoMEGA M2 1 set (1 right unit, 1 left unit)



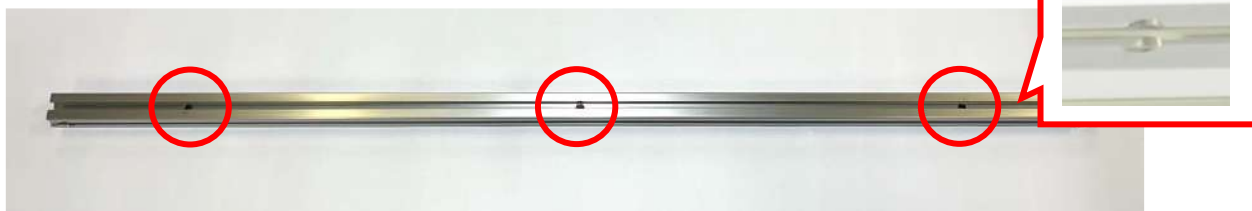
2.2 Coupling frames (750 [mm]) 3 units



2.3 Battery mounting frames (380 [mm]) 2 units



2.4 Loading frame 1 (1121 [mm]) 2 pieces



2.5 Loading frame 2 (1121 [mm]) 2 pieces



2.6 Loading frame 3 (120 [mm]) 4 pieces



2.7 Loading frame 4 (750 [mm]) 2 pieces



2.8 Electrical enclosure (with mounting aluminum frame) 1 [unit]



2.9 Emergency stop switch 2 [units]



2.10 Battery 1 [unit]



2.11 Battery securing belt 2 [pieces]



2.12 Battery Harness 1[unit]



2.13 Top Plate 3[pieces]



2.14 KOPROPO MC-8 1[unit]



2.15 Small Bag Parts



① Connecting Screws (M6-12)
64 [pcs]



② Connecting Nuts
72 [items]



③ Connecting Brackets
32 [items]



④ Bed Frame Connecting Screws
(M8-30)
14 [pcs]



⑤ Emergency Stop Switch Fixing Screws
(M6-16)
4 [pcs]



⑥ Frame Cover Screws
(M6-10)
8 [pcs]



⑦ Frame Cover L
2 [pcs]



⑧ Frame Cover R
2 [pcs]



⑨ Small Frame Cover
4 [pcs]

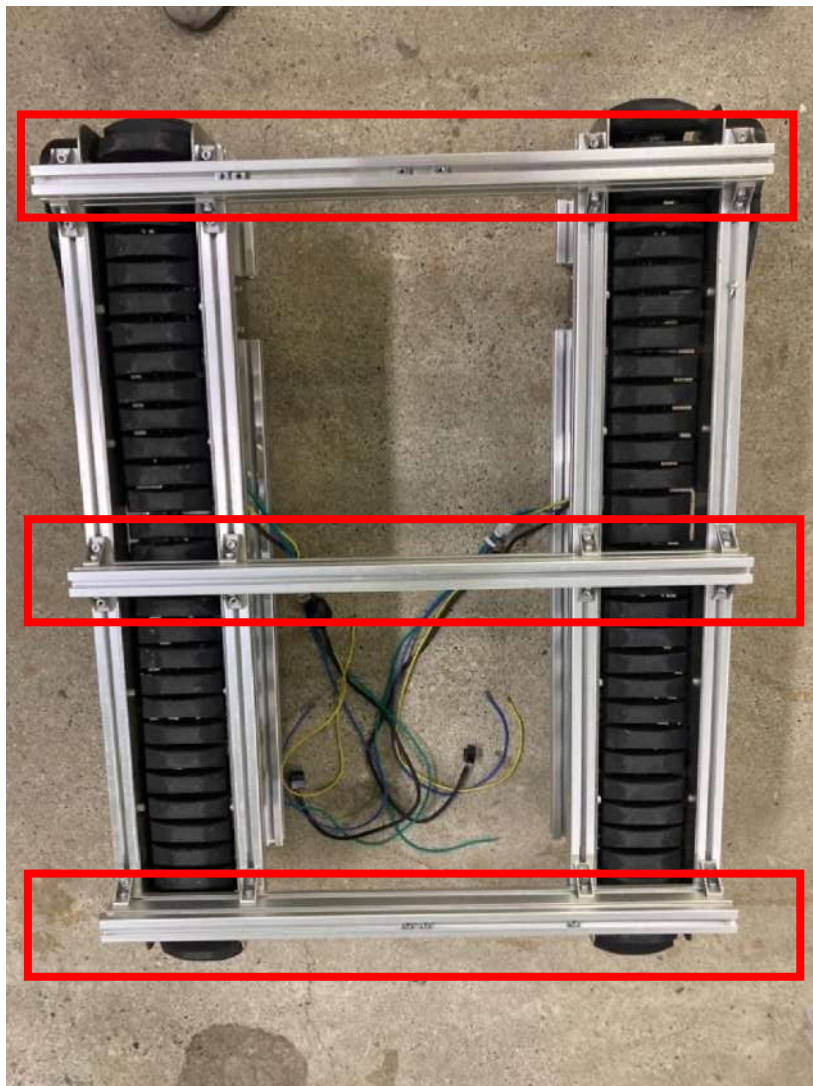


⑭ Screws for fixing the tabletop (M6-18)

12 [pcs]

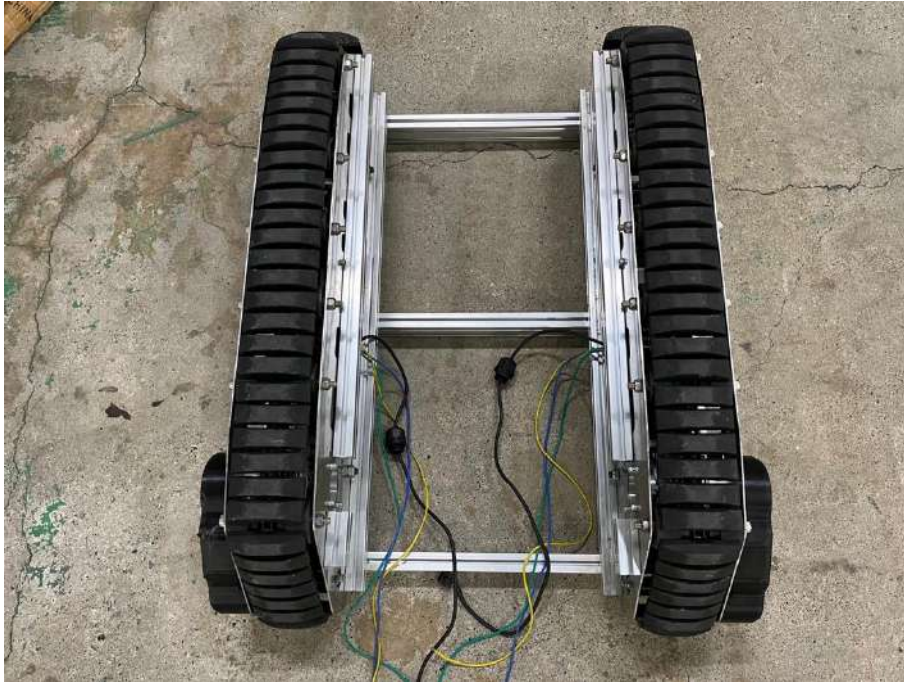
3 Assembly

[1] Attach the connecting frame (2.2) to CuGoMEGA M2 (2.1). For fastening, use the connecting screws (2.15⑭), the connecting nuts (2.15⑮), and the connecting brackets (2.15⑯).



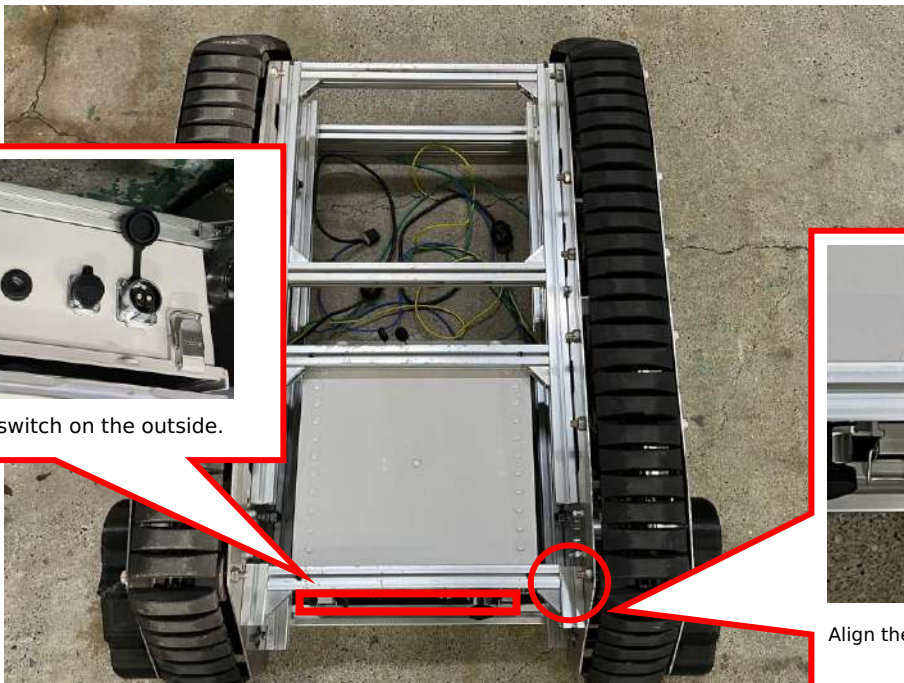
[2] Please invert CuGoMEGA M2.

Please make sure to work with at least two people.



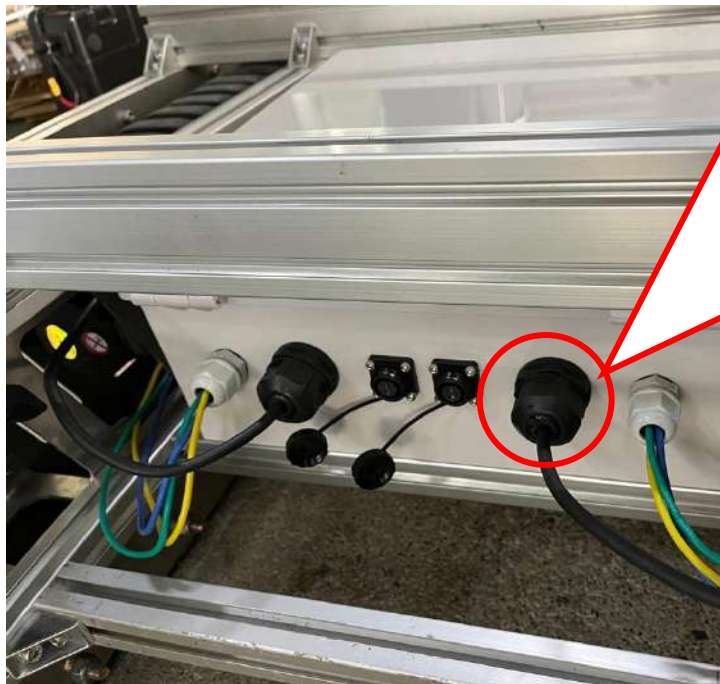
[3] Secure the battery mounting frame (2.3) and the electrical box (2.9) to CuGoMEGA M2. Fixation

Use the connecting screws (2.15①), connecting nuts (2.15②), and connecting brackets (2.15③). Please pay attention to the orientation of the brackets and the electrical box during installation.

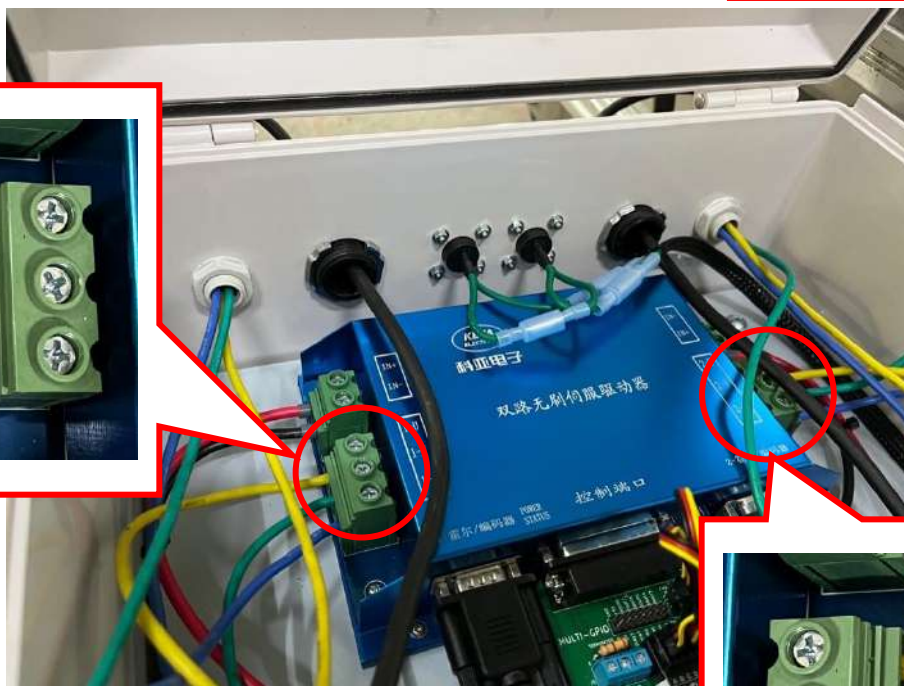


[4] Please invert CuGoMEGA M2 again.

[5] Pull the CuGoMEGA M2 harness (green, yellow, blue) and the encoder harness into the electrical box, and motor driver and connect them to the motor driver.



Remove the terminal cover, then thread them through the hole.



Please pay attention to the colors.

[6] Please connect the encoder harness.

Please connect it so you do not mix up left and right.



[7] Fix the battery in the battery mounting frame (2.3) using the battery fastening belt.



[8] Connect the battery harness (2.12) to the battery and route the wiring along the right side of the battery and electrical box.

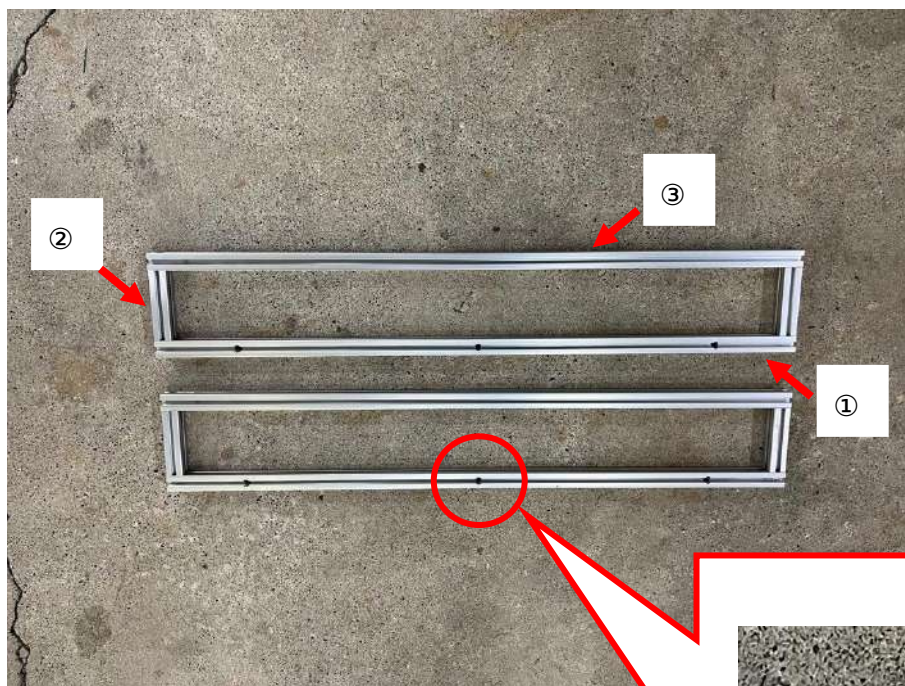


battery side



electrical box side

[9] Assemble frames for the cargo bed: ① (2.4), ② (2.5), and ③ (2.6) to produce two sets of cargo bed side panels.



The frame with holes is below.

[10] Use the cargo bed frame connection screws (2.15⑦) for fixing.

Connect the cargo bed side crafted at [11] [9] using the cargo bed frame ④ (2.7). Cargo bed frame ④
Please insert two connect nuts (2.15②) at the top of (2.7) in advance.



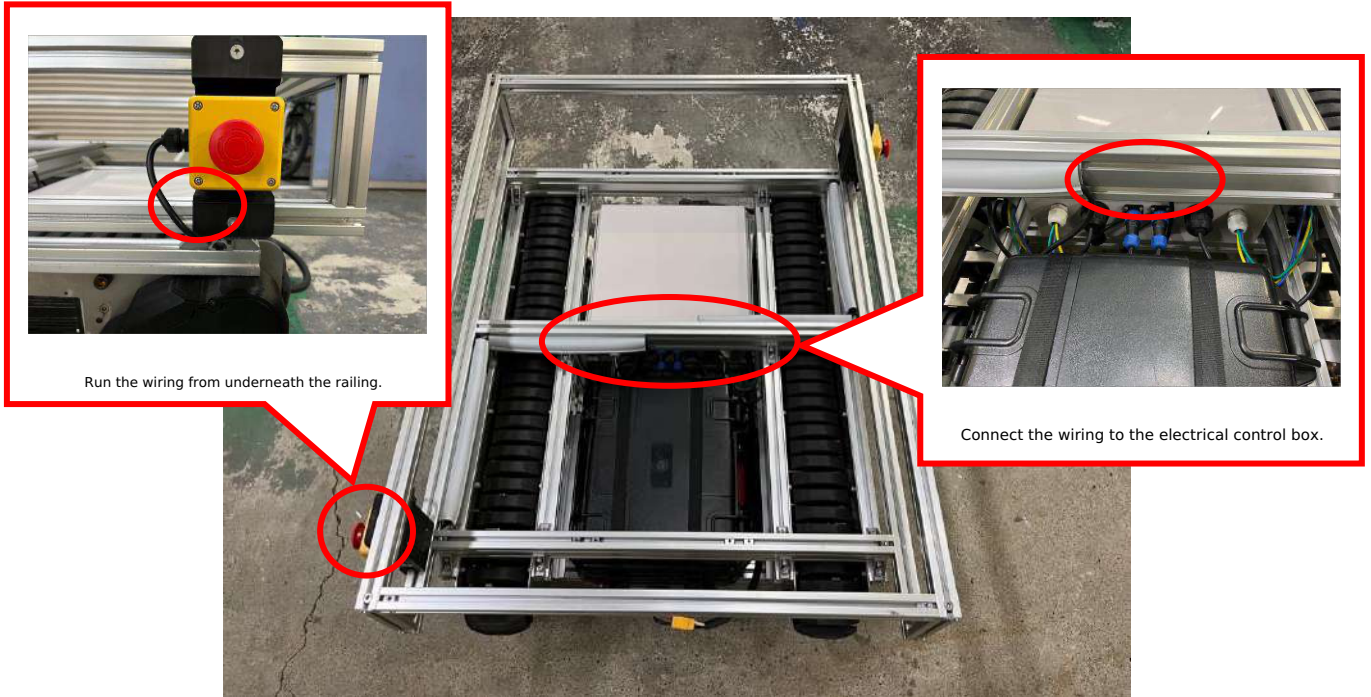
Please install the
frame nuts.

[12] Attach the cargo bed to the CuGoMEGA M2. For fixation, use the cargo bed frame connection screws (2.15④)
Please use.



[13] Wire and secure 2 emergency stop switches. Use coupling nuts (2.15②) for securing, the emergency stop splitting bolts (2.15⑦) for fixing.

*Please install ensuring the harness does not interfere with the top plate.



[14] Fix the frame cover (2.15⑥□⑧) to the corners of the cargo bed. Use frame cover screws (2.15⑤) for fastening.



[15] Secure the top board. Use the top board fixing screws (2.15®) for securing.



[16] Please connect the battery harness to the electrical box.



3.1 Power On

- 1) Use the switch on the electrical enclosure to turn on the main power.

The blue LED on the electrical enclosure will light up.



- 2) Release the emergency stop switch.

The yellow LED on the electrical enclosure will light up.



Hold down the controller power button to turn it on.

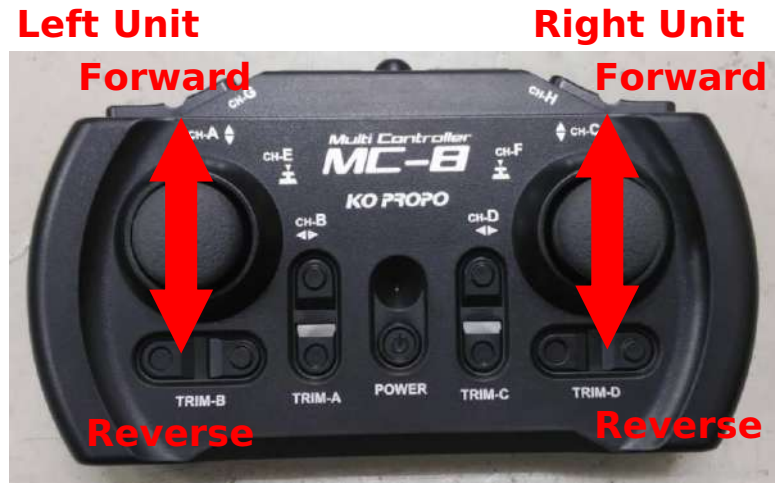
Be careful not to operate the joystick when turning on the power.



3.2 Operating Method

In the initial setup, the left and right sticks operate the left and right units respectively in 'Left-Right Independent Operation Mode'. The operation is as shown below, allowing turning by adjusting the speed difference between left and right.

By changing the wiring connections and motor controller settings, it can be switched to a '1-stick operation mode' that can be operated with a single stick (P.21-22)



4 Change settings

Connect the motor controller inside the electrical box to the PC using a USB-RS232 conversion cable, and use the configuration software to set your preferred settings. The following introduces changing the crawler's acceleration/deceleration and the method of operation.

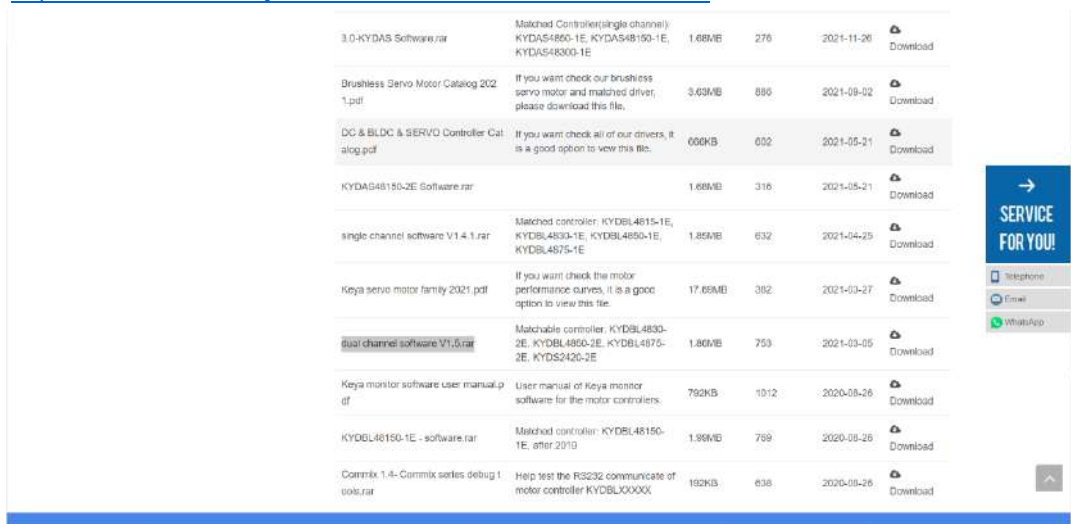
- “hY USB-RS232 変換ケーブル” (USB-Serial RS232 Cable)

<https://www.unitek-products.com/ja/products/usb-to-serial-adapter>

- “双チャンネルソフトウェア” (Dual Channel Software)

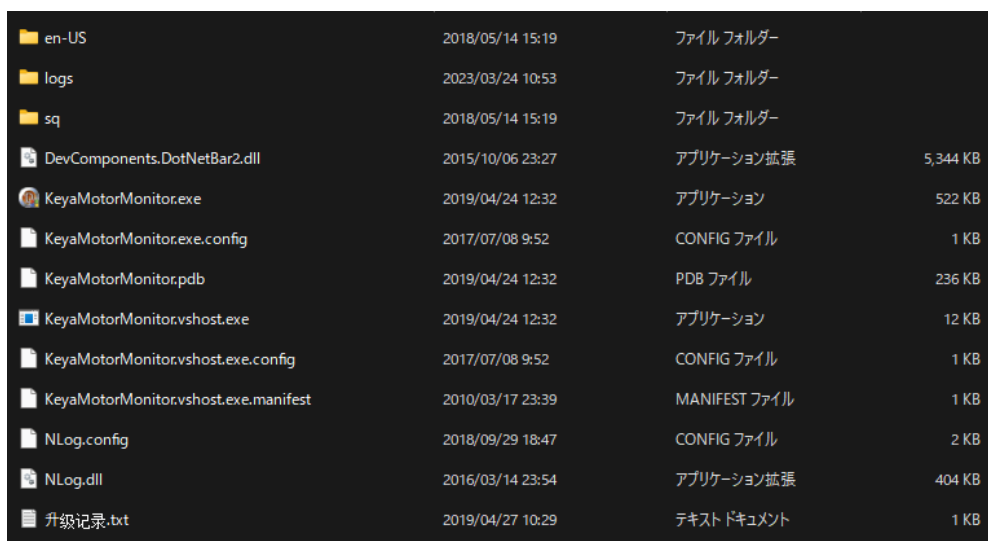
1. From the URL below, please download “dual channel software V1.5.rar.”

<http://www.dcmotorkeya.com/download.html>



3.0-KYDAS Software.rar	Matched Controller(single channel): KYDAS4800-1E, KYDAS48150-1E, KYDAS48300-1E	1.68MB	276	2021-11-26	Download
Brushless Servo Motor Catalog 2021.pdf	If you want check our brushless servo motor and matched driver, please download this file.	3.63MB	886	2021-09-02	Download
DC & BLDC & SERVO Controller Catalog.pdf	If you want check all of our drivers, it is a good option to view this file.	609KB	602	2021-05-21	Download
KYDAS48150-2E Software.rar		1.68MB	316	2021-05-21	Download
single channel software V1.4.1.rar	Matched controller: KYDBL4815-1E, KYDBL4830-1E, KYDBL4850-1E, KYDBL4875-1E	1.85MB	632	2021-04-25	Download
Keya servo motor family 2021.pdf	If you want check the motor performance curves, it is a good option to view this file.	17.65MB	382	2021-03-27	Download
dual channel software V1.5.rar	Matchable controller: KYDBL4830-2E, KYDBL4860-2E, KYDBL4875-2E, KYDS2420-2E	1.80MB	753	2021-03-05	Download
Keya monitor software user manual.pdf	User manual of Keya monitor software for the motor controllers.	792KB	1012	2020-08-26	Download
KYDBL48150-1E - software.rar	Matched controller: KYDBL48150-1E, after 2019	1.96MB	789	2020-08-26	Download
Commix 1.4-Commix series debug tool.rar	Help test the RS232 communicate of motor controller KYDBLXXXX	192KB	638	2020-08-26	Download

2. Unzip what you downloaded and check if there is a file named “KeyaMotorMonitor.exe” in the folder. Please confirm. That file will be the configuration software.

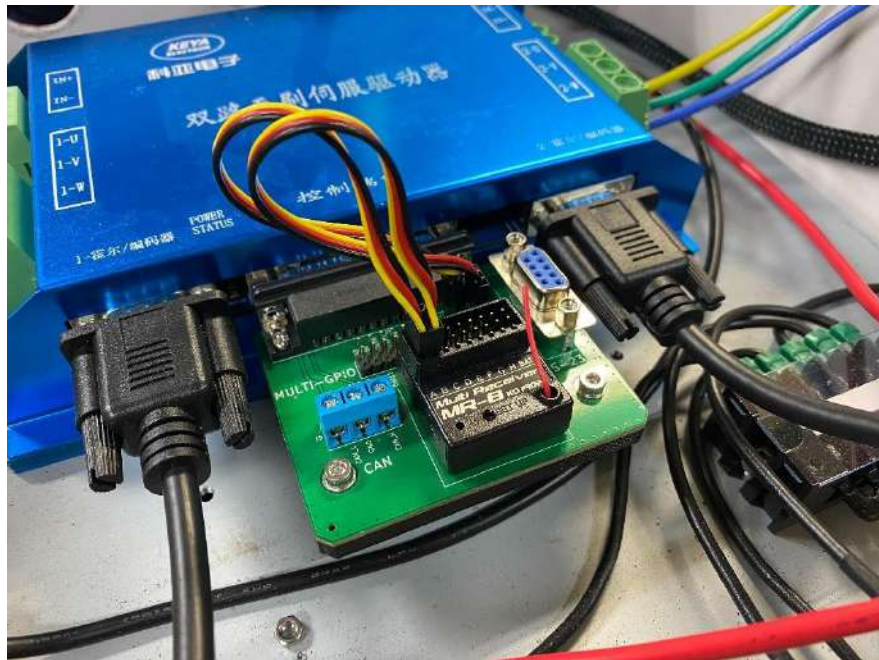


en-US	2018/05/14 15:19	ファイル フォルダー	
logs	2023/03/24 10:53	ファイル フォルダー	
sq	2018/05/14 15:19	ファイル フォルダー	
DevComponents.DotNetBar2.dll	2015/10/06 23:27	アプリケーション 拡張	5,344 KB
KeyaMotorMonitor.exe	2019/04/24 12:32	アプリケーション	522 KB
KeyaMotorMonitor.exe.config	2017/07/08 9:52	CONFIG ファイル	1 KB
KeyaMotorMonitor.pdb	2019/04/24 12:32	PDB ファイル	236 KB
KeyaMotorMonitor.vshost.exe	2019/04/24 12:32	アプリケーション	12 KB
KeyaMotorMonitor.vshost.exe.config	2017/07/08 9:52	CONFIG ファイル	1 KB
KeyaMotorMonitor.vshost.exe.manifest	2010/03/17 23:39	MANIFEST ファイル	1 KB
NLog.config	2018/09/29 18:47	CONFIG ファイル	2 KB
NLog.dll	2016/03/14 23:54	アプリケーション 拡張	404 KB
升级记录.txt	2019/04/27 10:29	テキスト ドキュメント	1 KB

4.1 Parameter changes for crawler acceleration/deceleration

You can configure the crawler's acceleration/deceleration. Lowering the acceleration/deceleration reduces shocks during starting and stopping.

1. Please connect the motor controller to the PC using a USB-RS232 conversion cable.

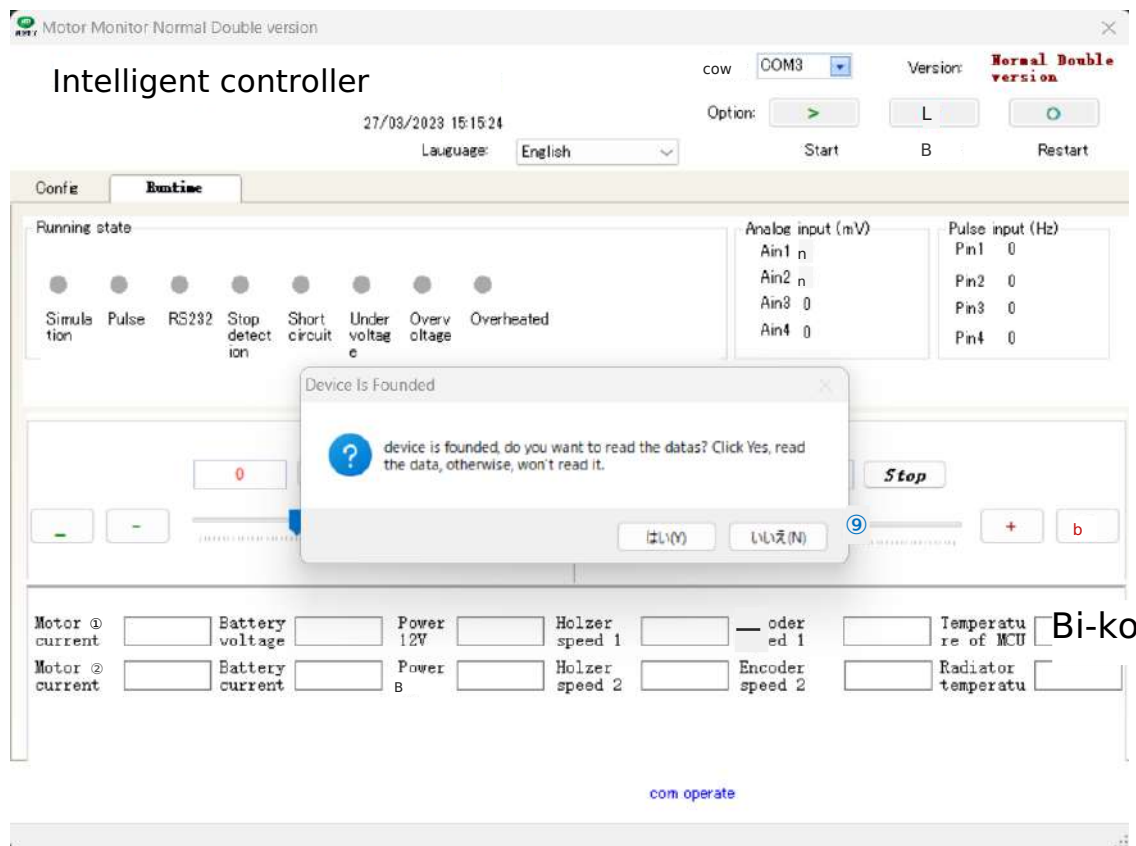


2. Turn on the power and confirm that the LED lights up.



3. Start KeyaMotorMonitor.exe and confirm that the controller is connected to the PC.

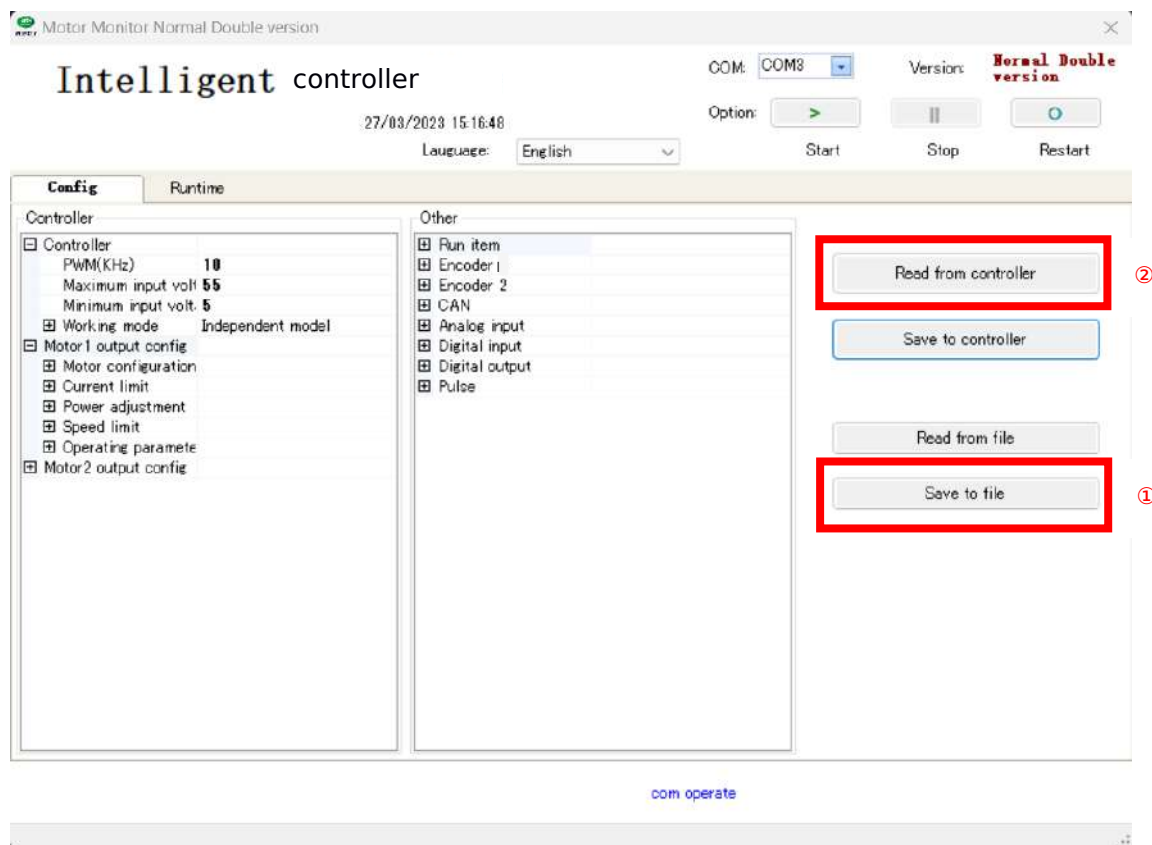
When confirmed, a window like the following will appear, please click “Yes (Y)”. If it does not appear, please check that the connection is properly established.



4. Save the initial settings written to the motor controller to the PC.

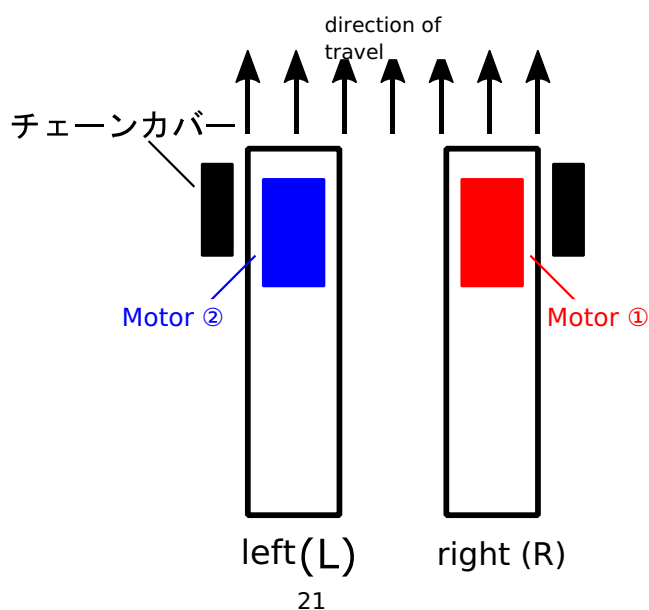
First, click “Read from controller” to read the already written initial settings. Then click “Save to file” to save to the PC.

Note: After changing settings, you may need to revert to the initial settings. Therefore, please save the initial settings. If you accidentally erase the initial settings data, please contact us.



5. Change the settings values.

Change the parameters of 'Soft start (RPM/S)' and 'Soft stop (RPM/S)' for Moto1 and Motor2. In the initial settings, both are 3000 (RPM/S). Motor 1 is the right motor,



Motor2 is assigned to the left motor. Typically the same value is set for both sides.

Controller

Contoler

PWMKHz) ⑩

Maximum input vol ⑤⑤

Minimum input vol ⑤

Working mode frdependent model

Motor① output confie

Motor configuration

Current limit

Power adjustment

Speed limit

Max speed rpm 3000

Soft start (RPM) 3000

Soft stop (RPM) 03000

Number of pole- ⑤

Operating parameter

Motor② output confie

Motor configuration

Current limit

Power adjustment

Speed limit

Max speed rpm 3000

Soft start (RPM) 3000

Soft stop (RPM) 3000

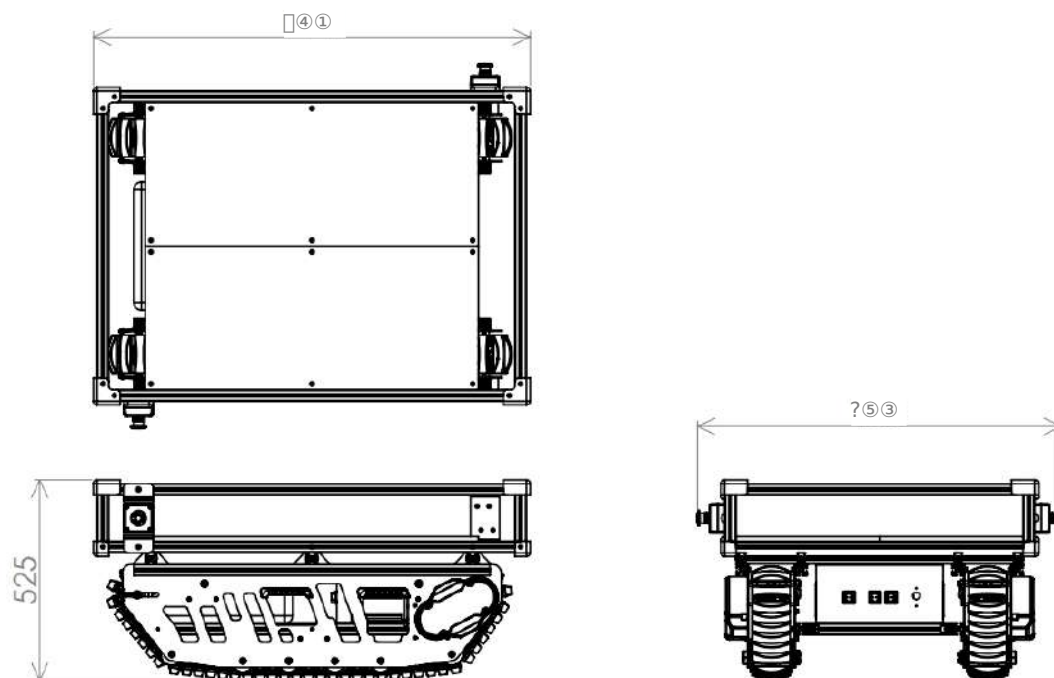
Number of pole- ⑤

Operating parameter

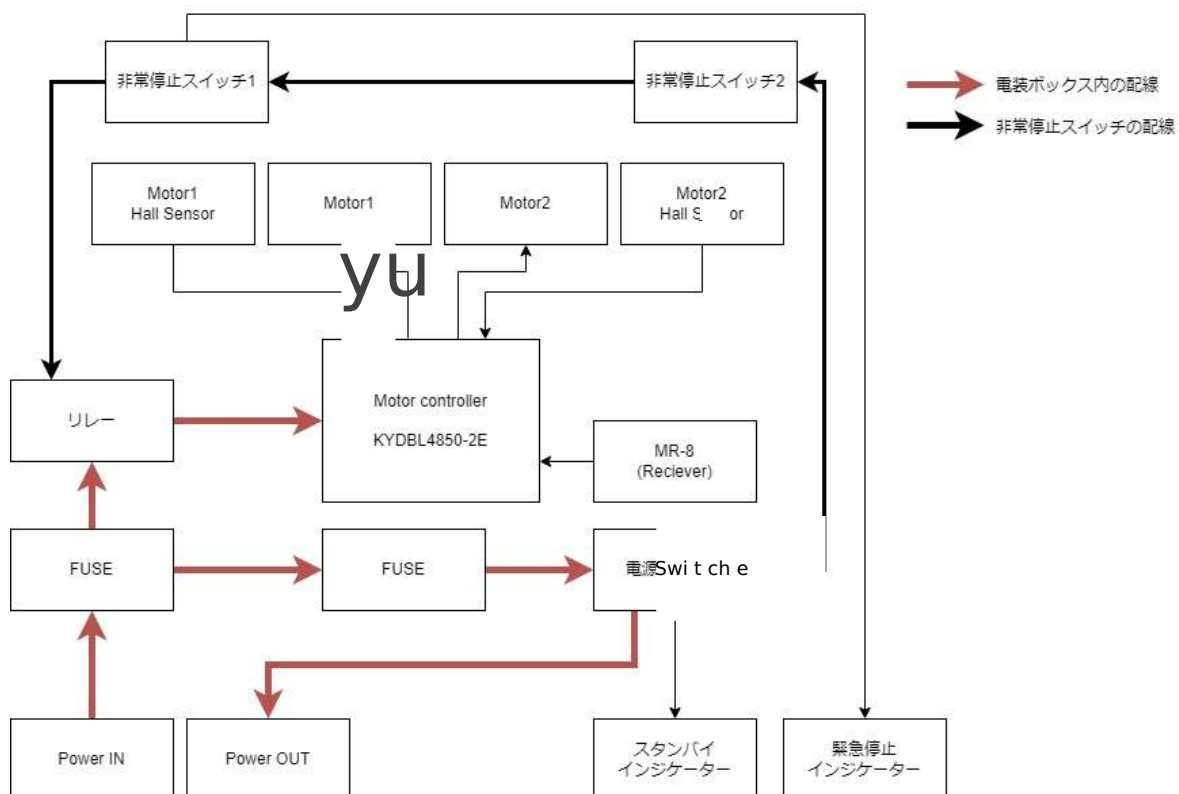
6. Click “Save to controller” to save the settings to the motor controller.

This completes the acceleration/deceleration parameter changes. If there are any issues, please contact us.

5 Dimension Diagram



6 Control Unit Wiring Diagram



7 Contact Us

Contact form: Phone: <https://cuborex.com/contactform/>
070-8535-9595



Note